

CAPSTONE PROJECT REPORT

Best Buy Automation | Selenium Testing Report

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Superset ID	4959462
Project Type	Capstone Project
Project Name	Best Buy Automation Testing using Selenium PyTest and Behave BDD Framework
Domain	E-Commerce
Module Title	Deal of the Day

PROJECT DESCRIPTION

This project automates the Best Buy web application using Selenium WebDriver with Python. The automation framework covers the Deal of the Day module on the Best Buy e-commerce platform, validating deal navigation, product selection, cart operations, checkout flows, and email validation using both Selenium PyTest and Behave BDD frameworks.

The framework is designed using the Page Object Model (POM) architecture to improve maintainability and reusability. Screenshots, logs, Allure reports, and reusable utilities are integrated to support execution tracking and debugging. The project includes both positive and negative testing scenarios using industry-standard automation testing practices.

PART 1 — SELENIUM PYTEST FRAMEWORK

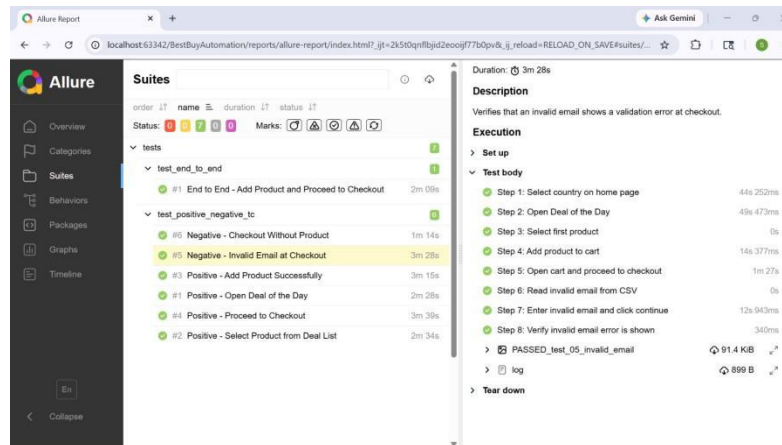


Fig-1: Selenium PyTest Framework

1.1 Test Strategy

The Selenium PyTest framework automates the Best Buy Deal of the Day module using Selenium WebDriver with Python PyTest. The framework validates deal page navigation, product selection, cart operations, checkout flow, and email validation including negative cases for invalid email input and empty cart checkout. It supports End-to-End, Positive, and Negative testing with CSV-based data-driven test data.

Allure Reports are integrated for professional execution reporting with screenshot evidence and detailed execution logs.

Tools Used:

- Selenium WebDriver
- PyTest
- Allure Reports
- Python Logging Module
- Screenshot Utility

1.2 Test Environment

Operating System	Windows 11
IDE	PyCharm
Browser	Google Chrome
Framework	Selenium + PyTest
Design Pattern	Page Object Model (POM)
Reporting Tool	Allure Reports
Language	Python 3.12.2

1.3 Framework Architecture

```
BestBuyAutomation/
├── conftest.py
├── requirements.txt
├── pytest.ini
├── config/
│   └── config.properties
├── data/
│   └── test_data.csv
├── pages/
│   ├── base_page.py
│   ├── home_page.py
│   ├── deal_of_day_page.py
│   ├── cart_page.py
│   ├── checkout_page.py
│   └── product_page.py
├── tests/
│   ├── test_positive_negative_tc.py
│   └── test_end_to_end.py
├── utils/
│   ├── logger.py
│   ├── screenshot_util.py
│   ├── csv_reader.py
│   └── config_reader.py
├── logs/
│   └── automation.log
├── reports/
│   ├── screenshots/
│   ├── allure-results/
│   └── allure-report/
```

1.4 Test Execution

End-to-End Test

The End-to-End test validates the complete Best Buy Deal of the Day workflow. Starting from the home page, the test selects the user's country, navigates to the Deal of the Day page, selects the first product, adds it to the cart, navigates to the cart, proceeds to checkout, reads a valid email from the CSV data file, enters the email, and asserts that the checkout or sign-in page is reached successfully.

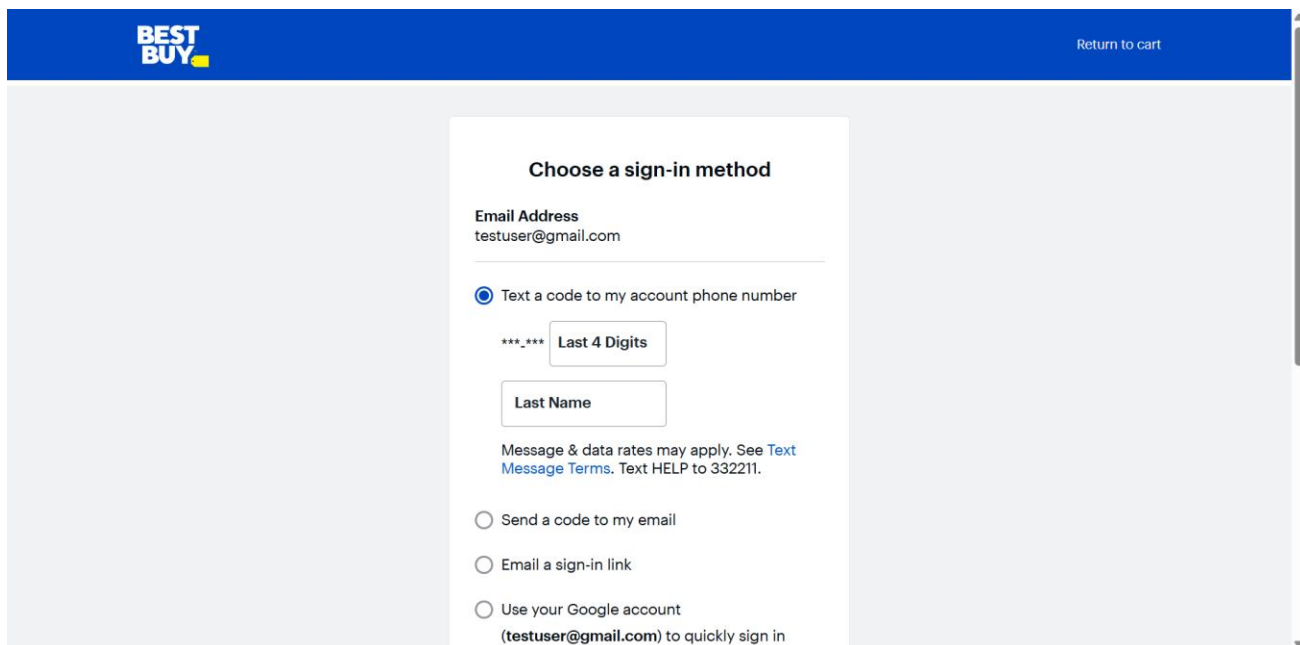
Positive and Negative Tests

The positive test cases validate: opening the Deal of the Day page, selecting a product from the deal list, adding a product to the cart, and proceeding to checkout. The negative test cases validate: invalid email input at checkout displaying a validation error message, and navigating to the cart without adding any product detecting an empty cart state.

1.5 Test Case Summary — Selenium PyTest

S.No	Test Type	Test Scenario	Expected Result	Status
1	End-to-End	End to End - Add Product and Proceed to Checkout (Deal of Day → Cart → Checkout → Email)	Checkout/sign-in page reached after email entry	Passed
2	Positive	Open Deal of the Day	Deal of the Day page displayed successfully	Passed
3	Positive	Select Product from Deal List	Product details page displayed after selection	Passed
4	Positive	Add Product Successfully to Cart	Cart is not empty after adding product	Passed
5	Positive	Proceed to Checkout	Checkout page opens after cart navigation	Passed
6	Negative	Invalid Email at Checkout (from CSV)	Validation error displayed for invalid email	Passed
7	Negative	Checkout Without Product (Empty Cart)	Empty cart detected and confirmed	Passed

1.6 Test Outputs & Screenshots



The screenshot displays the Best Buy checkout process. At the top, the Best Buy logo is on the left, and a "Return to cart" link is on the right. The main content area is a white box titled "Choose a sign-in method". It shows the email address "testuser@gmail.com" entered in a field. Below this, the option "Text a code to my account phone number" is selected with a radio button. This option includes a field for "Last 4 Digits" (displayed as "****") and a "Last Name" field. A message states: "Message & data rates may apply. See [Text Message Terms](#). Text HELP to 332211." Below this message are three unselected radio button options: "Send a code to my email", "Email a sign-in link", and "Use your Google account (testuser@gmail.com) to quickly sign in".

Fig-2 [PyTest] End-to-End Test — Checkout Page Reached

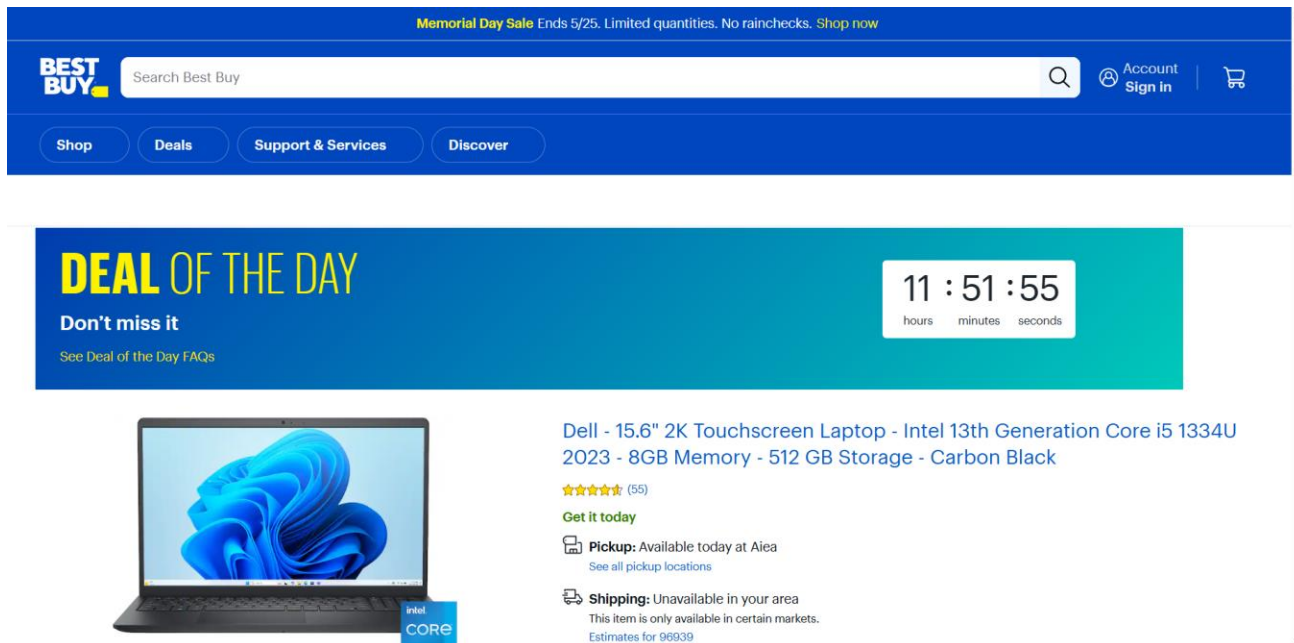


Fig-3 [PyTest] TC01 — Deal of the Day Page Opened

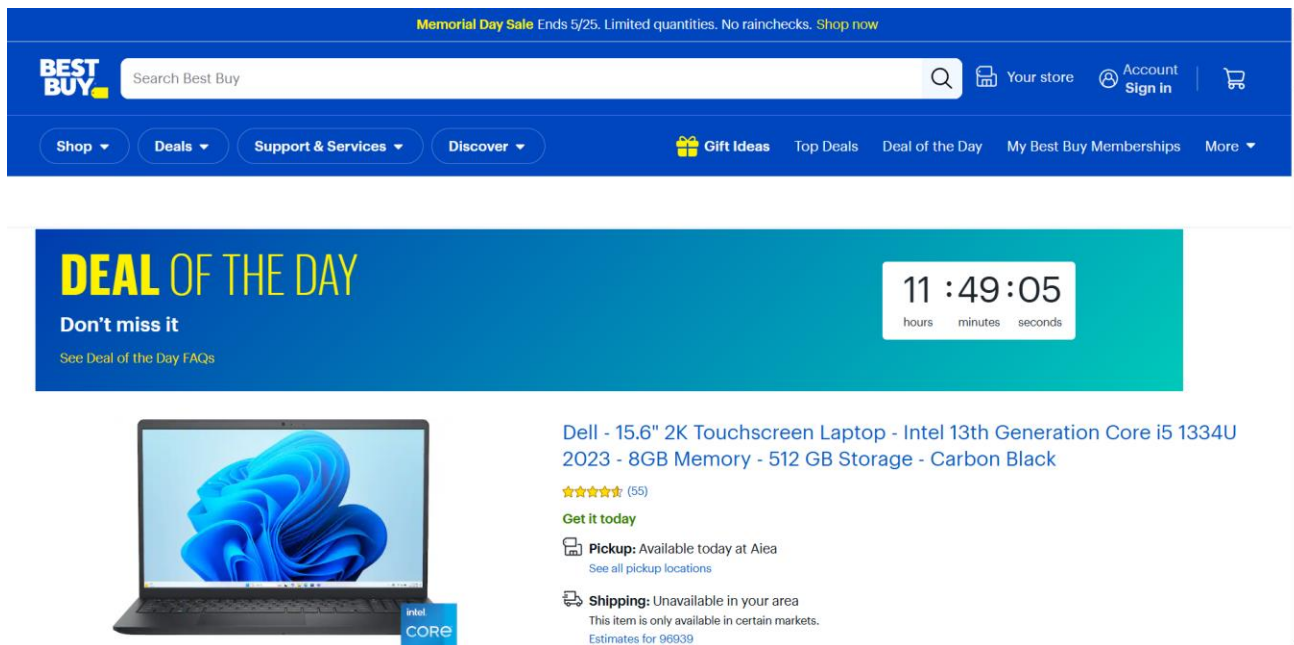


Fig-4 [PyTest] TC02 — Product Selected from Deal List

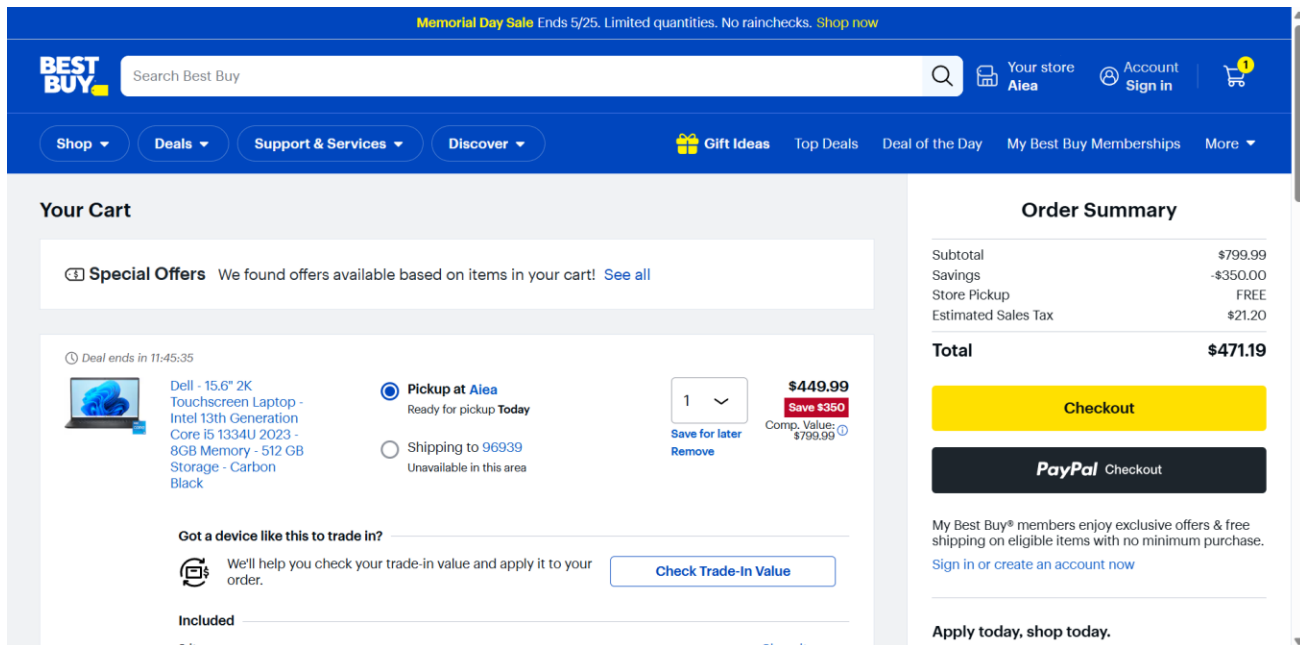


Fig-5 [PyTest] TC03 — Product Added to Cart

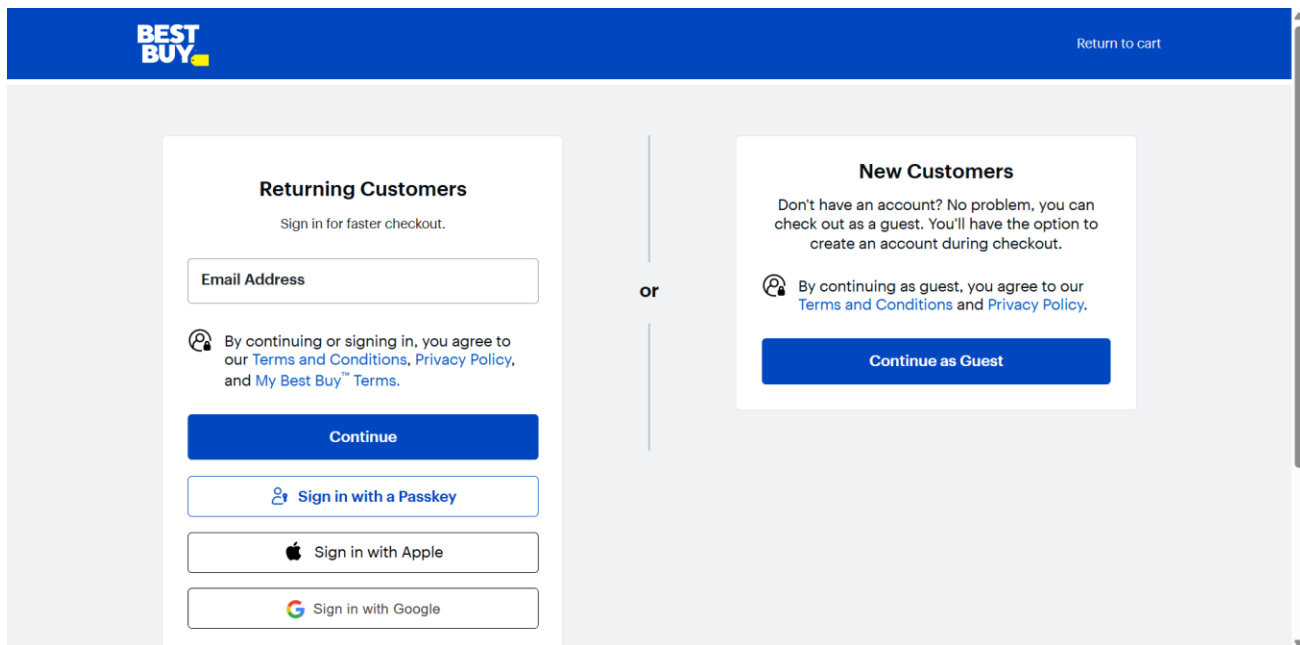


Fig-6 [PyTest] TC04 — Checkout Page Displayed

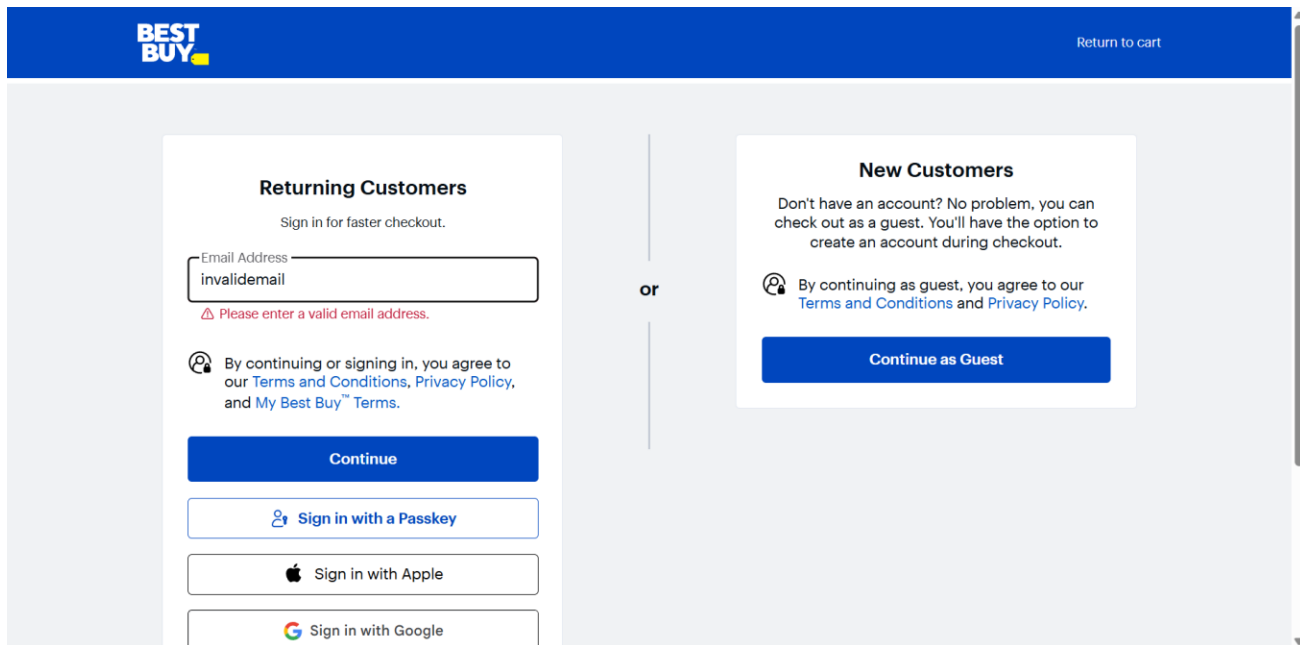


Fig-7 [PyTest] TC05 — Invalid Email Validation Error

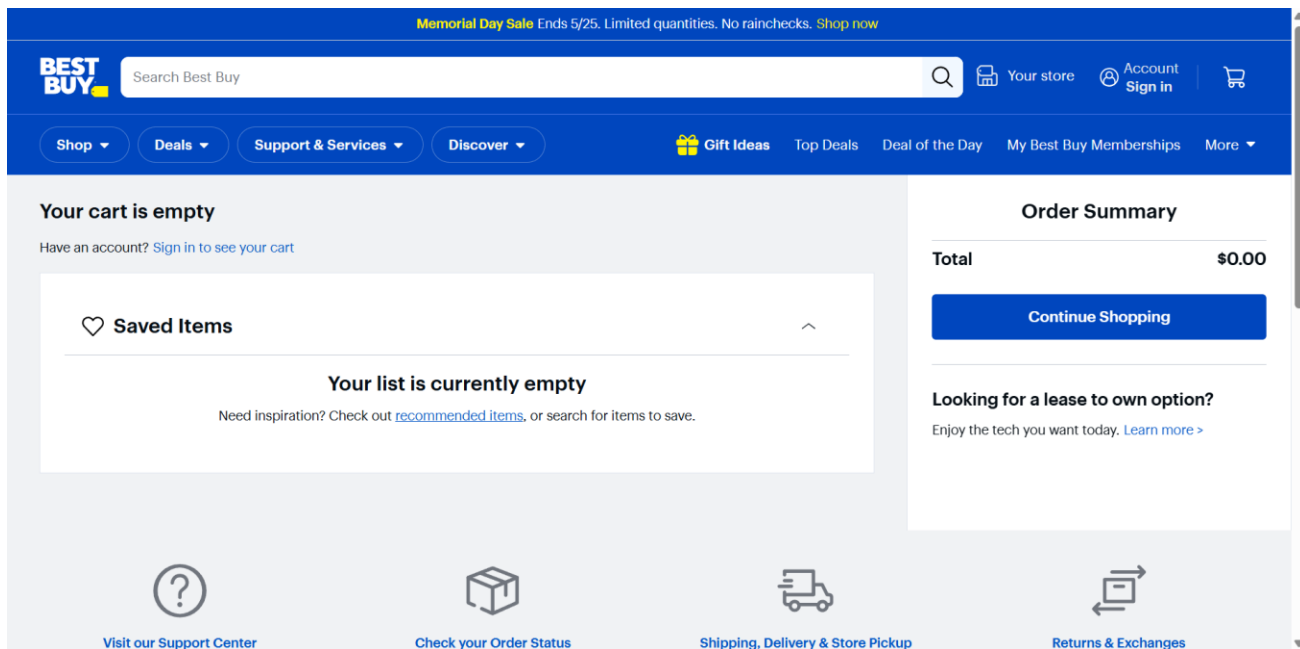


Fig-8 [PyTest] TC06 — Empty Cart Detected

1.7 Allure Report — Selenium PyTest

Allure Reports are integrated with the Selenium PyTest framework to generate professional execution reports. The reports display passed status, capture screenshots at key execution stages, and attach per-test logs for debugging.

Memorial Day Sale Ends 5/25. Limited quantities. No rainchecks. [Shop now](#)

BEST BUY Search Best Buy   Account Sign In 

[Shop](#) [Deals](#) [Support & Services](#) [Discover](#)

DEAL OF THE DAY

Don't miss it

[See Deal of the Day FAQs](#)

11 : 51 : 55
hours minutes seconds



Dell - 15.6" 2K Touchscreen Laptop - Intel 13th Generation Core i5 1334U 2023 - 8GB Memory - 512 GB Storage - Carbon Black

★★★★★ (55)

[Get it today](#)

 **Pickup:** Available today at Aiea
[See all pickup locations](#)

 **Shipping:** Unavailable in your area
This item is only available in certain markets.
[Estimates for 96939](#)

Fig-9 [PyTest Allure] TC01 — Open Deal of the Day

Memorial Day Sale Ends 5/25. Limited quantities. No rainchecks. [Shop now](#)

BEST BUY Search Best Buy   Your store  Account Sign In 

[Shop](#) [Deals](#) [Support & Services](#) [Discover](#) [Gift Ideas](#) [Top Deals](#) [Deal of the Day](#) [My Best Buy Memberships](#) [More](#)

DEAL OF THE DAY

Don't miss it

[See Deal of the Day FAQs](#)

11 : 49 : 05
hours minutes seconds



Dell - 15.6" 2K Touchscreen Laptop - Intel 13th Generation Core i5 1334U 2023 - 8GB Memory - 512 GB Storage - Carbon Black

★★★★★ (55)

[Get it today](#)

 **Pickup:** Available today at Aiea
[See all pickup locations](#)

 **Shipping:** Unavailable in your area
This item is only available in certain markets.
[Estimates for 96939](#)

Fig-10 [PyTest Allure] TC02 — Select Product from Deal List

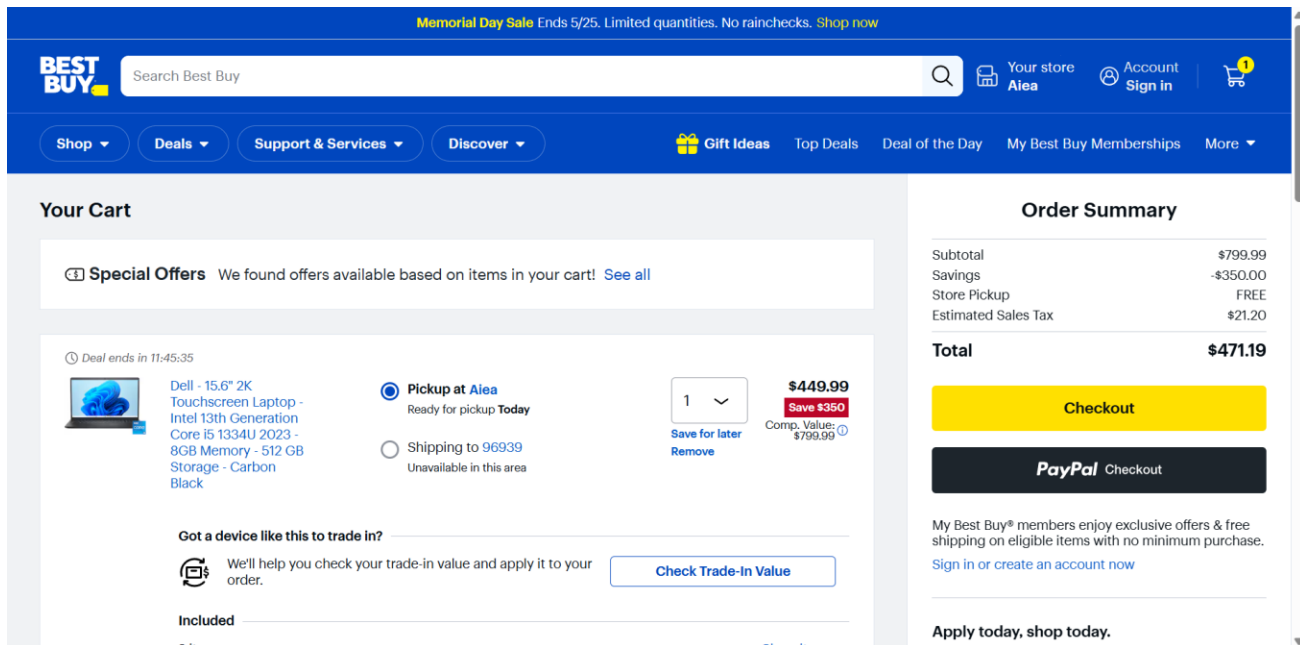


Fig-11 [PyTest Allure] TC03 — Add Product to Cart

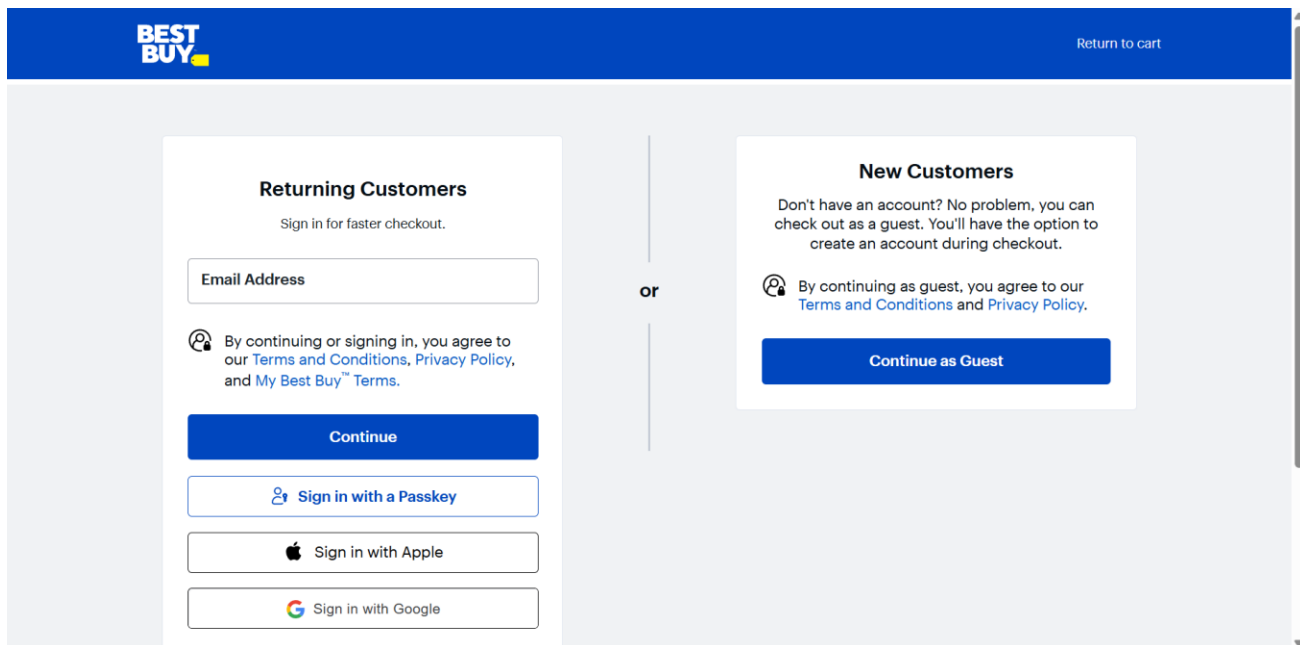


Fig-12 [PyTest Allure] TC04 — Proceed to Checkout

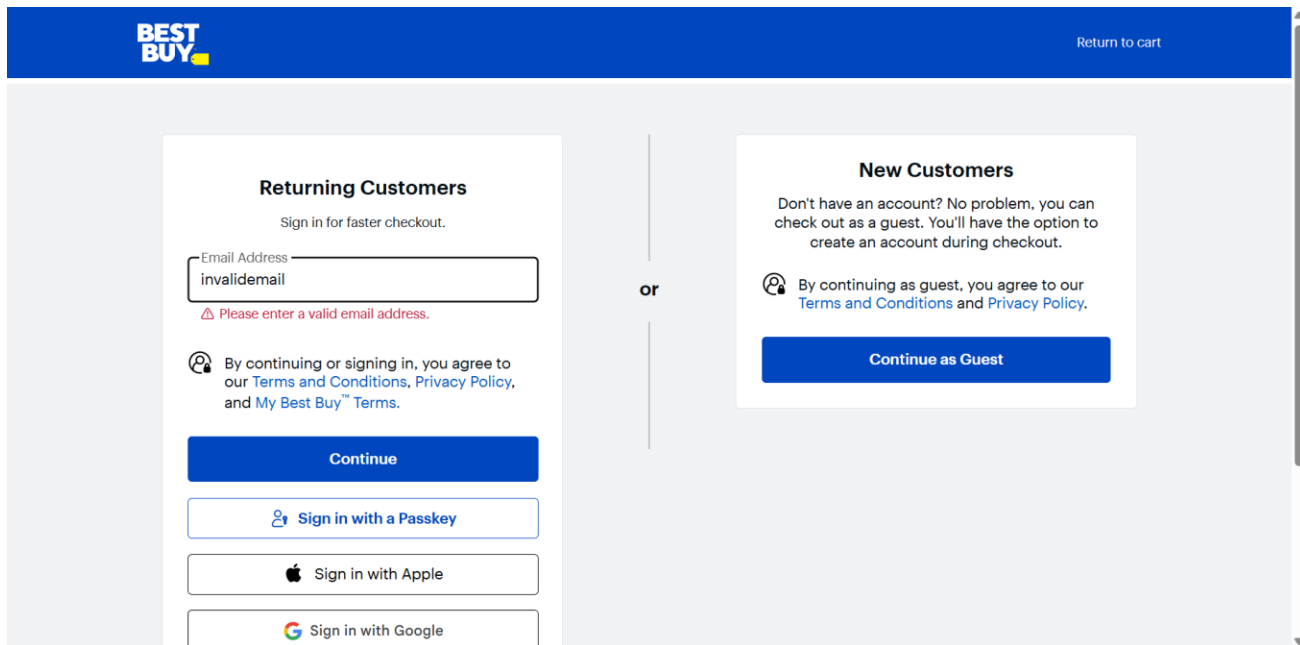


Fig-13 [PyTest Allure] TC05 — Invalid Email at Checkout

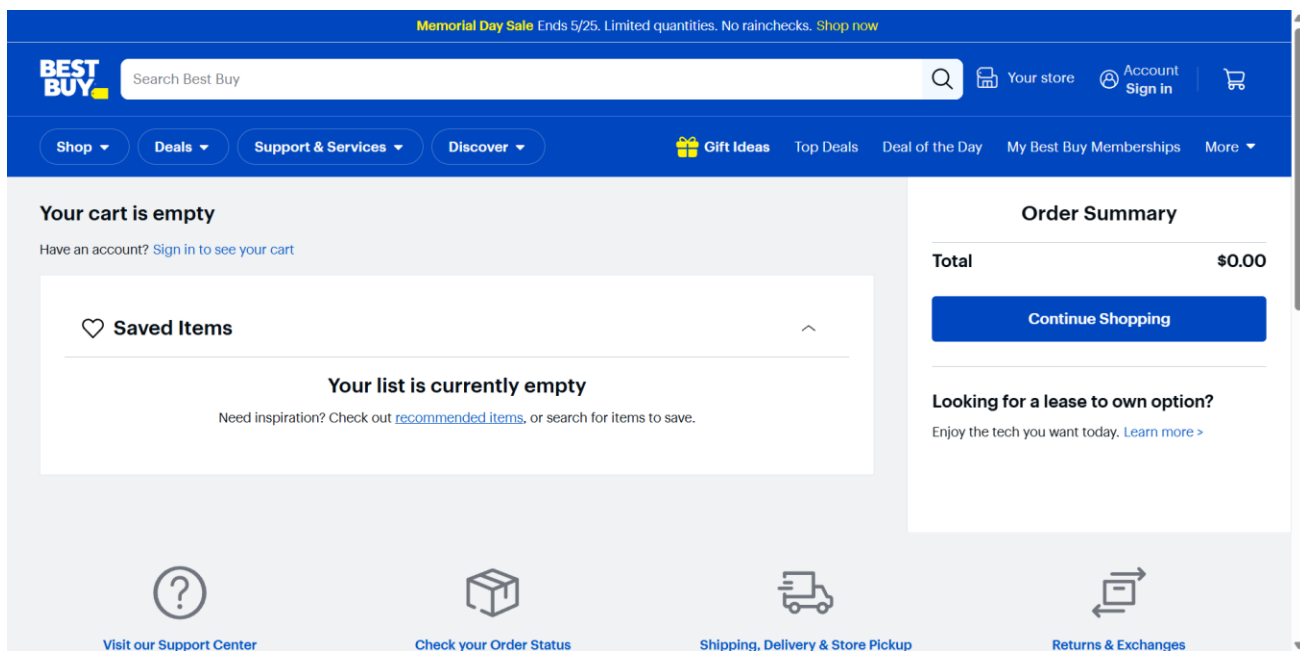


Fig-14 [PyTest Allure] TC06 — Checkout Without Product

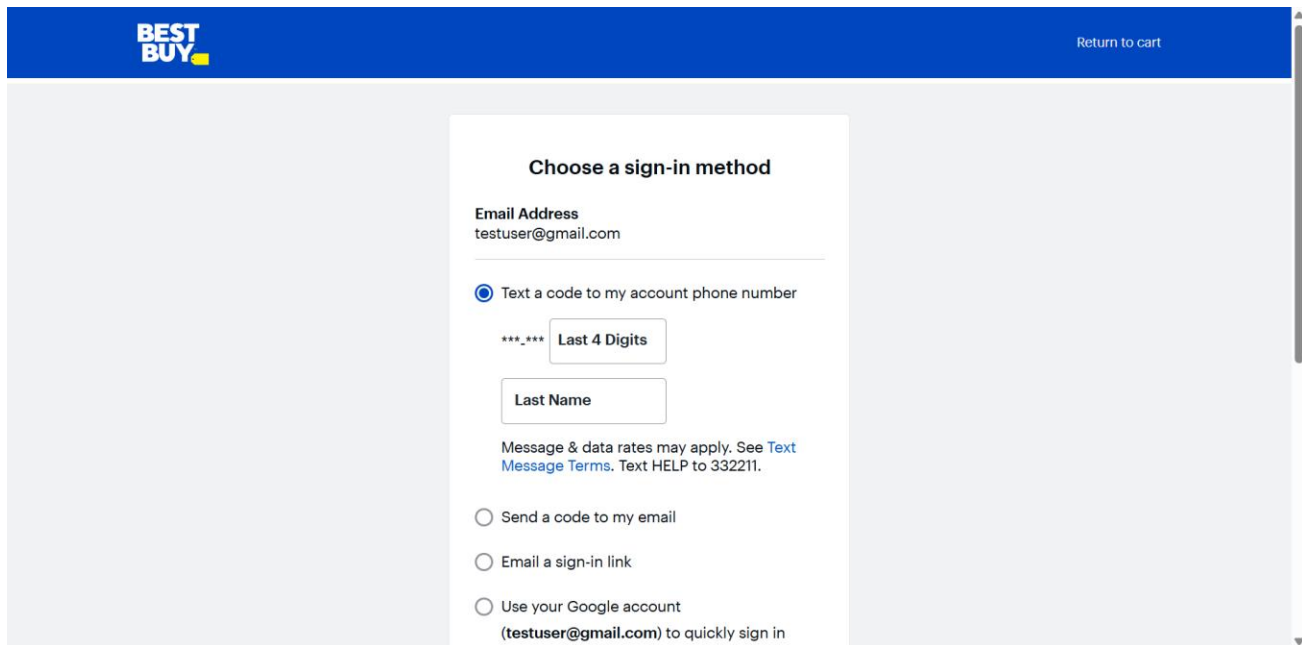


Fig-15 [PyTest Allure] End-to-End Test Passed

1.8 Test Report Summary — Selenium PyTest

Total Tests Executed	7
Tests Passed	7
Tests Failed	0
Automation Accuracy	100%

1.9 Conclusion — Selenium PyTest

The Selenium PyTest automation framework successfully validates the Best Buy Deal of the Day module using End-to-End, Positive, and Negative testing approaches. The framework leverages Selenium WebDriver, PyTest, Page Object Model, Allure Reporting, Logging Framework, and Data-Driven Testing with CSV. All 7 test cases executed successfully with professional reporting, screenshot capture, and detailed execution logs demonstrating automation accuracy of 100%.

PART 2 — SELENIUM BDD BEHAVE FRAMEWORK

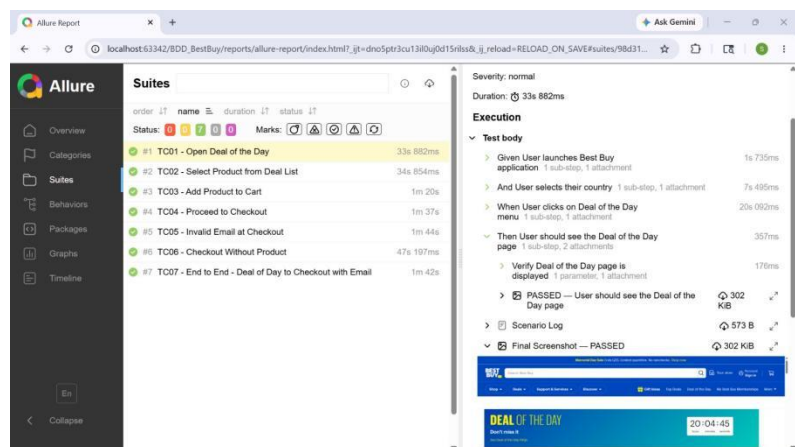


Fig-16: Selenium BDD Behave Framework

2.1 Test Strategy

The Selenium Behave BDD framework automates the Best Buy Deal of the Day module using Behavior Driven Development methodology. Test scenarios are written in Gherkin syntax using Given-When-Then structure, making them readable by both technical and non-technical stakeholders. The framework validates the full deal-to-checkout flow expressed as business-readable feature files.

Allure Reports are integrated with the Behave formatter to generate professional execution reports with per-scenario screenshot evidence and isolated log attachments.

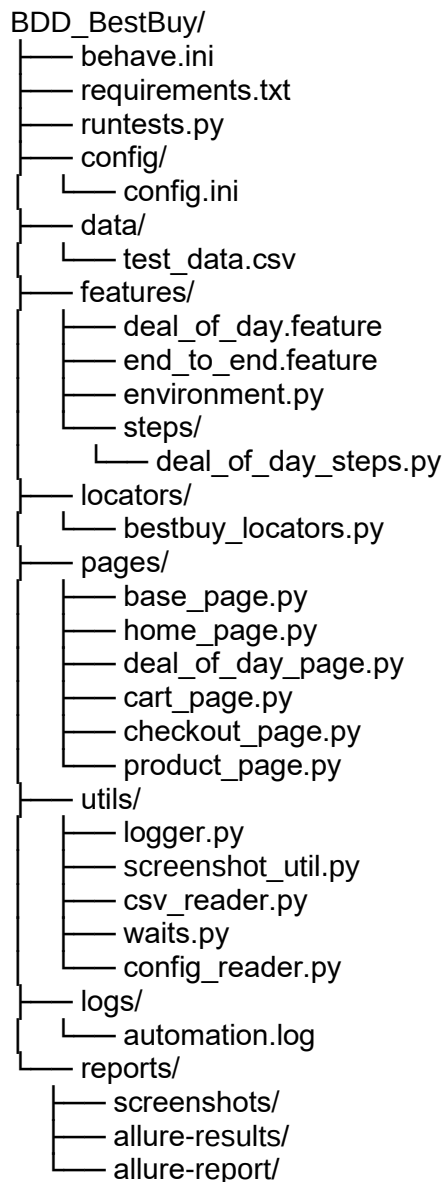
Tools Used:

- Selenium WebDriver
- Behave BDD Framework
- Allure Reports (allure-behave)
- Python Logging Module

2.2 Test Environment

Operating System	Windows 11
IDE	PyCharm / VS Code
Browser	Google Chrome
Framework	Selenium + Behave BDD
Design Pattern	Page Object Model (POM)
Reporting Tool	Allure Reports
Test Approach	Behavior Driven Development (BDD)
Language	Python 3.12.2

2.3 Framework Architecture



2.4 BDD Implementation — Feature Files

The framework follows Behavior Driven Development using Feature Files, Step Definition Files, Gherkin Syntax, and Given-When-Then structure. A Background step initialises the browser and navigates to Best Buy before each scenario. Tags (@positive, @negative, @e2e, @smoke) are used to categorise and selectively run scenarios.

Feature: deal_of_day.feature (TC01–TC06)

Feature: Best Buy Deal of the Day - Functional Tests (Positive & Negative) Background:
 Given User launches Best Buy application And User selects their country @positive @smoke
 Scenario: TC01 - Open Deal of the Day When User clicks on Deal of the Day menu Then User
 should see the Deal of the Day page @positive Scenario: TC02 - Select Product from Deal List
 When User clicks on Deal of the Day menu And User selects the first product from the deal list
 Then User should see the product details page @positive Scenario: TC03 - Add Product to Cart
 When User clicks on Deal of the Day menu And User selects the first product from the deal list
 And User clicks on Add to Cart button And User clicks on Cart icon Then User should see
 the cart is not empty @positive Scenario: TC04 - Proceed to Checkout When User clicks on
 Deal of the Day menu ... And User clicks on Checkout button Then User should see the
 checkout page @negative Scenario: TC05 - Invalid Email at Checkout ... When User
 enters an invalid email "invalidemail@@@.com" And User clicks on Continue button Then User
 should see an invalid email error message @negative Scenario: TC06 - Checkout Without Product
 When User opens the cart directly without adding a product Then User should see the cart is
 empty

Feature: end_to_end.feature (TC07)

Feature: Best Buy Deal of the Day - End to End Tests Background: Given User launches Best
 Buy application And User selects their country @e2e Scenario: TC07 - End to End - Deal of
 Day to Checkout with Email When User clicks on Deal of the Day menu And User selects the
 first product from the deal list And User clicks on Add to Cart button And User clicks on
 Cart icon And User clicks on Checkout button Then User should see the checkout page
 When User enters email address "testuser@example.com" And User clicks on Continue button
 Then User should proceed to the next checkout step

2.5 Test Execution

End-to-End Scenario (TC07)

The End-to-End scenario validates the complete Best Buy Deal of the Day to checkout flow using BDD methodology. The Background step launches the browser and navigates to Best Buy. The scenario then clicks on Deal of the Day, selects the first product, adds it to the cart, proceeds to checkout, enters a valid email address, and asserts that the next checkout step is reached successfully.

Positive Scenarios (TC01–TC04)

TC01 verifies that the Deal of the Day page loads successfully after clicking the menu. TC02 verifies that a product can be selected from the deal list and the product details page opens. TC03 adds the selected product to the cart and confirms the cart is not empty. TC04 continues through to checkout and verifies the checkout page is displayed.

Negative Scenarios (TC05–TC06)

TC05 enters an invalid email "invalidemail@@@.com" at checkout and verifies that a validation error message is displayed. TC06 opens the cart directly without adding any product and asserts that the cart is empty.

2.6 Test Case Summary — Selenium BDD Behave

S.No	Test Type	Test Scenario	Expected Result	Status
1	Positive / Smoke	TC01 - Open Deal of the Day	Deal of the Day page displayed successfully	Passed

S.No	Test Type	Test Scenario	Expected Result	Status
2	Positive	TC02 - Select Product from Deal List	Product details page displayed after selection	Passed
3	Positive	TC03 - Add Product to Cart	Cart is not empty after product added	Passed
4	Positive	TC04 - Proceed to Checkout	Checkout page opens successfully	Passed
5	Negative	TC05 - Invalid Email at Checkout ("invalidemail@@@.com")	Invalid email error message displayed	Passed
6	Negative	TC06 - Checkout Without Product	Cart is empty as expected	Passed
7	End-to-End	TC07 - End to End - Deal of Day to Checkout with Email	Next checkout step reached after email entry	Passed

2.7 Test Outputs & Screenshots

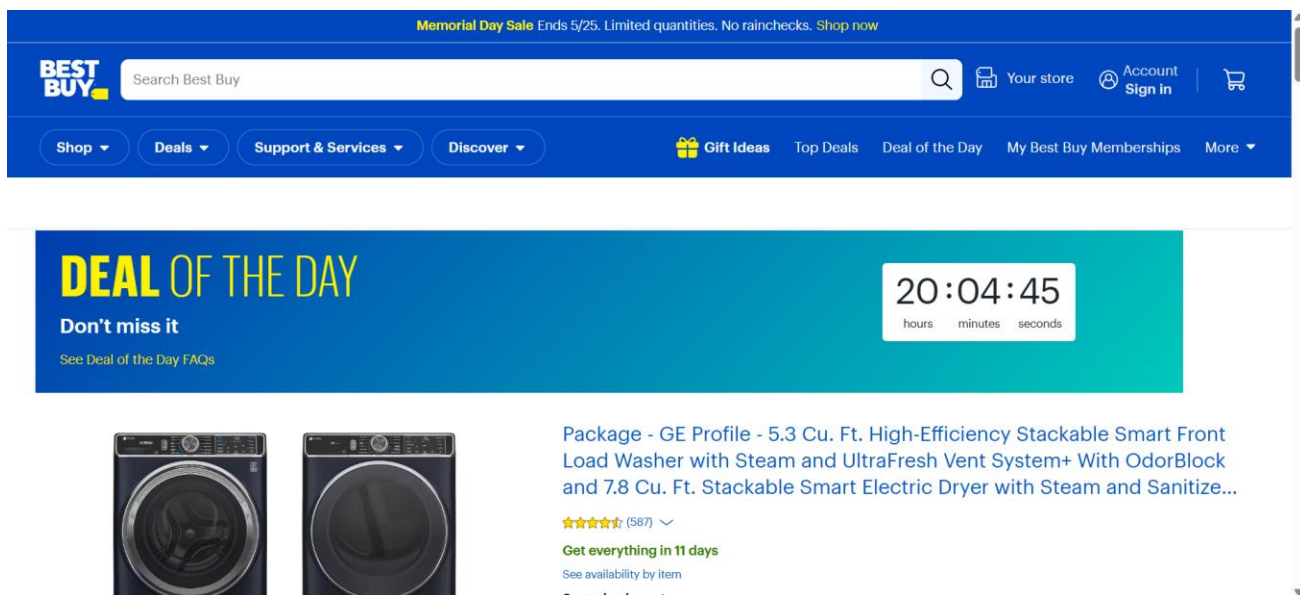


Fig-17 [BDD] TC01 — Deal of the Day Page Opened

Memorial Day Sale Ends 5/25. Limited quantities. No rainchecks. Shop now

BEST BUY

Search Best Buy

Your store

Account Sign in

Shop

Deals

Support & Services

Discover

Gift Ideas

Top Deals

Deal of the Day

My Best Buy Memberships

More

DEAL OF THE DAY

Don't miss it

See Deal of the Day FAQs

20:04:10

hoursminutesseconds

GE Profile - 5.3 Cu. Ft. High-Efficiency Stackable Smart Front Load Washer with Steam and UltraFresh Vent System+ With OdorBlock and 7.8 Cu. Ft. Stackable Smart Electric Dryer with Steam and Sanitize...

★★★★★ (587)

Get everything in 11 days

See availability by item

See price in cart

Package - GE Profile - 5.3 Cu. Ft. High-Efficiency Stackable Smart Front Load Washer with Steam and UltraFresh Vent System+ With OdorBlock and 7.8 Cu. Ft. Stackable Smart Electric Dryer with Steam and Sanitize...

Fig-18 [BDD] TC02 — Product Selected from Deal List

Memorial Day Sale Ends 5/25. Limited quantities. No rainchecks. Shop now

BEST BUY

Search Best Buy

Your store Aiea

Account Sign in

Shop

Deals

Support & Services

Discover

Gift Ideas

Top Deals

Deal of the Day

My Best Buy Memberships

More

Your Cart

GE Profile - 7.8 Cu. Ft. Stackable Smart Electric Dryer with Steam and Sanitize Cycle - Sapphire Blue

Pickup at Aiea

Ready for pickup Wed, Jun 3

Delivery to 96939

Unavailable in your area

1

Save for later

Remove

Memorial Day Deal

\$799.99

Save \$505

Comp. Value: \$1,304.99

Protection Plans

Standard Geek Squad Protection

See All Plans

Deal ends in 20:02:49

GE Profile - 5.3 Cu. Ft. High-Efficiency Stackable Smart Front

Pickup at Aiea

Ready for pickup Wed, Jun 3

1

Memorial Day Deal

\$649.99

Order Summary

Subtotal

\$2,609.98

Savings

-\$1,160.00

Delivery

FREE

Estimated Sales Tax

\$87.00

Total

\$1,536.98

Checkout

PayPal Checkout

My Best Buy® members enjoy exclusive offers & free shipping on eligible items with no minimum purchase. Sign in or create an account now

Fig-19 [BDD] TC03 — Product Added to Cart

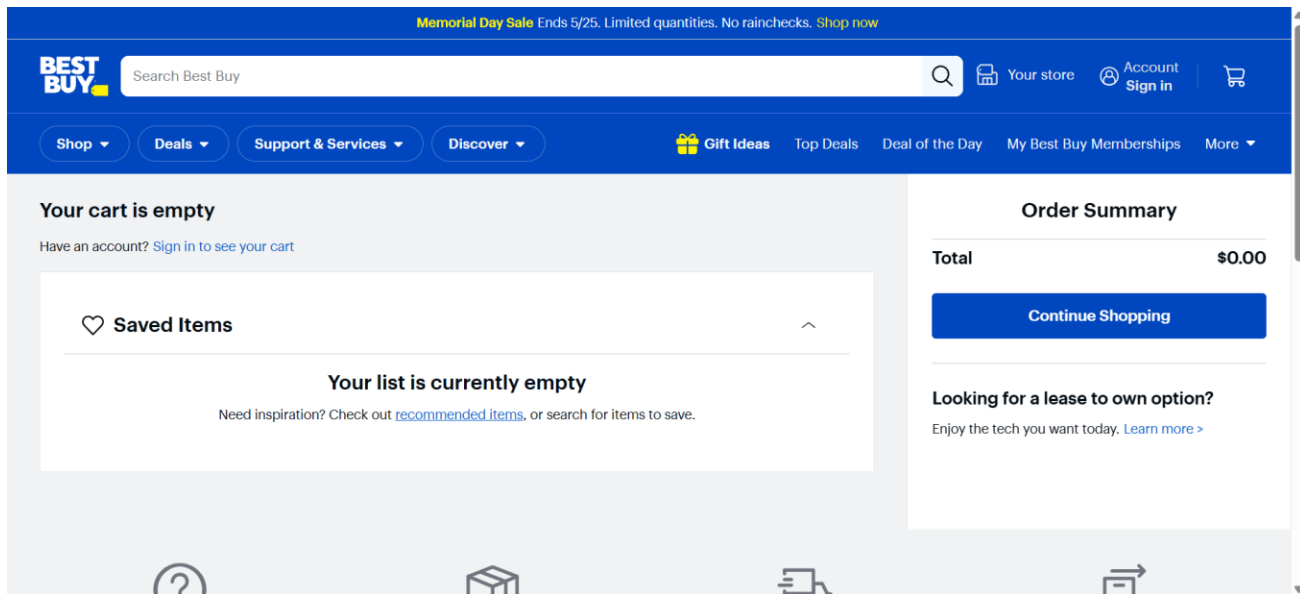


Fig-22 [BDD] TC06 — Empty Cart Detected

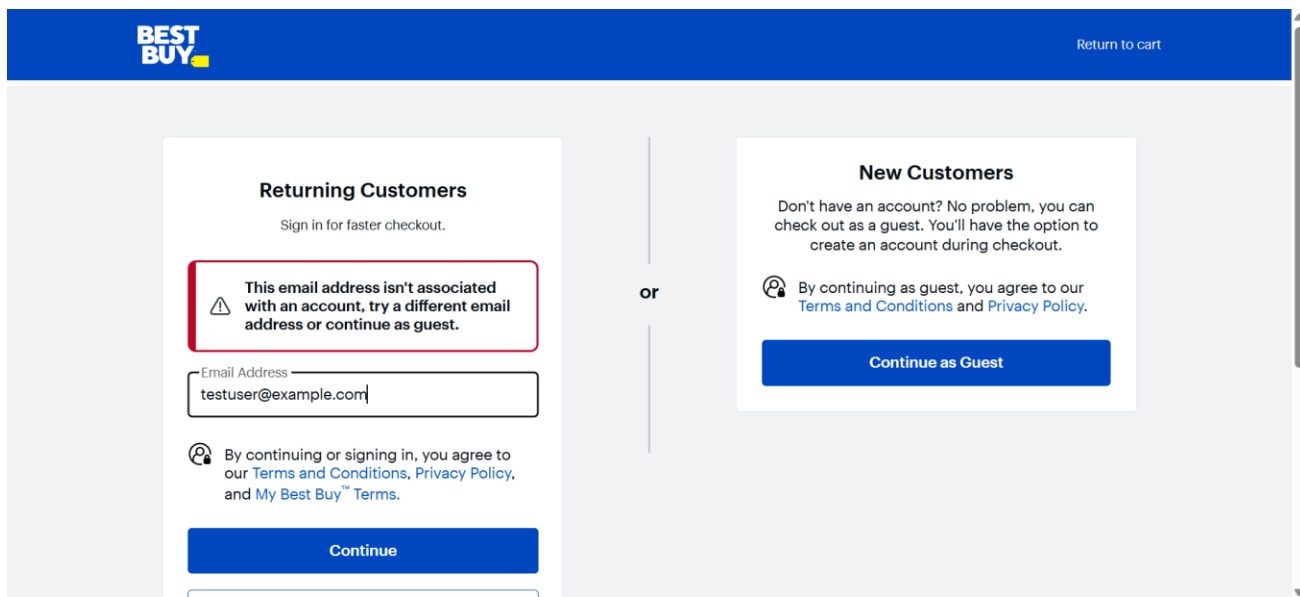


Fig-23 [BDD] TC07 — End-to-End: Next Checkout Step Reached

2.8 Allure Report — BDD Behave

Allure Reports are integrated with the Behave BDD framework. Screenshots are attached after every step, and a final screenshot is attached at the end of each scenario along with the full scenario log for debugging.

DEAL OF THE DAY

Don't miss it

[See Deal of the Day FAQs](#)

20:04:45
hours minutes seconds



Package - GE Profile - 5.3 Cu. Ft. High-Efficiency Stackable Smart Front Load Washer with Steam and UltraFresh Vent System+ With OdorBlock and 7.8 Cu. Ft. Stackable Smart Electric Dryer with Steam and Sanitize...

★★★★★ (587) 

Get everything in 11 days

[See availability by item](#)

[See price in cart](#)

Fig-24 [BDD Allure] TC01 — Open Deal of the Day

DEAL OF THE DAY

Don't miss it

[See Deal of the Day FAQs](#)

20:04:10
hours minutes seconds



Package - GE Profile - 5.3 Cu. Ft. High-Efficiency Stackable Smart Front Load Washer with Steam and UltraFresh Vent System+ With OdorBlock and 7.8 Cu. Ft. Stackable Smart Electric Dryer with Steam and Sanitize...

★★★★★ (587) 

Get everything in 11 days

[See availability by item](#)

[See price in cart](#)

Fig-25 [BDD Allure] TC02 — Select Product from Deal List (Final Screenshot)

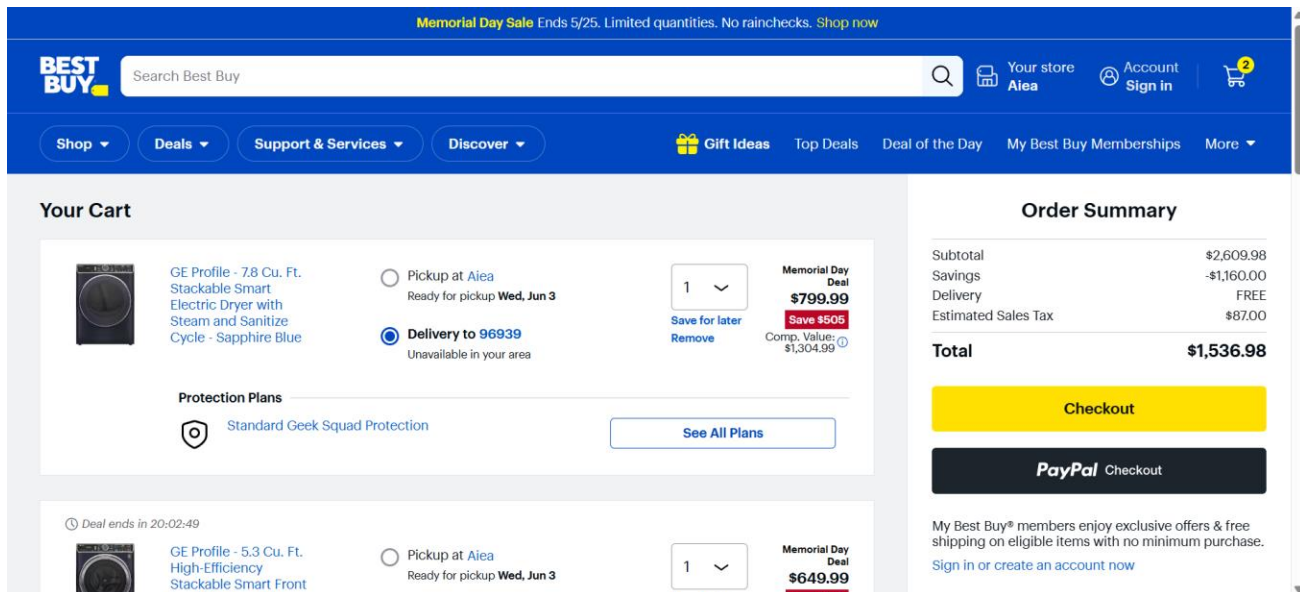


Fig-26 [BDD Allure] TC03 — Add Product to Cart (Final Screenshot)

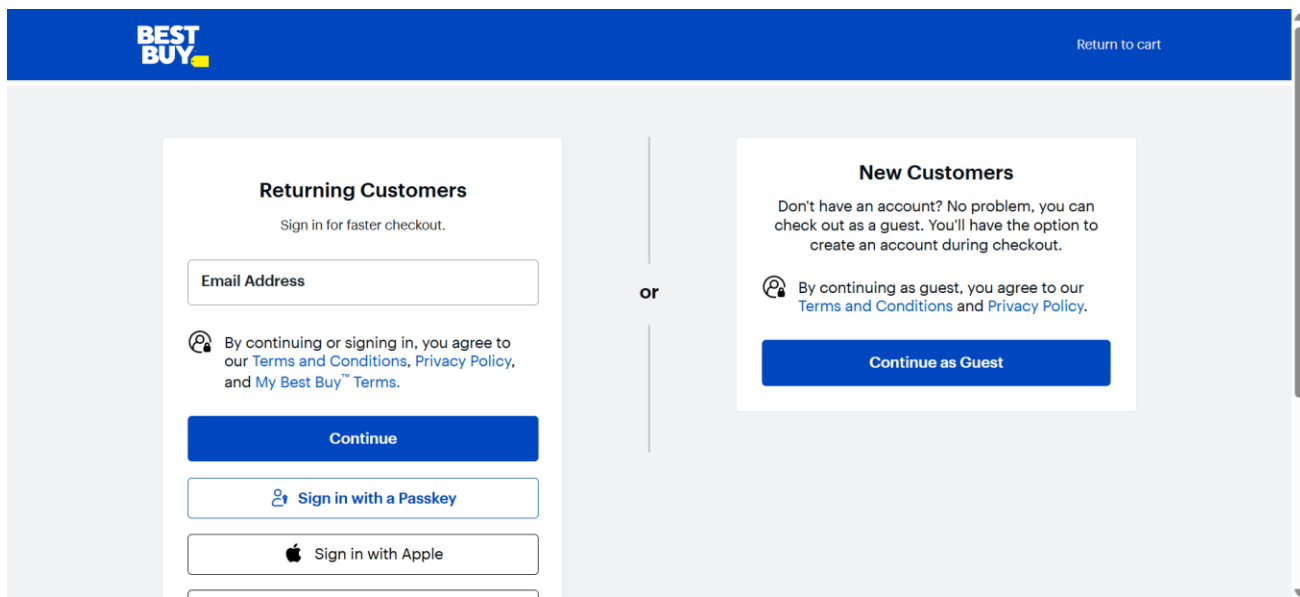


Fig-27 [BDD Allure] TC04 — Proceed to Checkout (Final Screenshot)

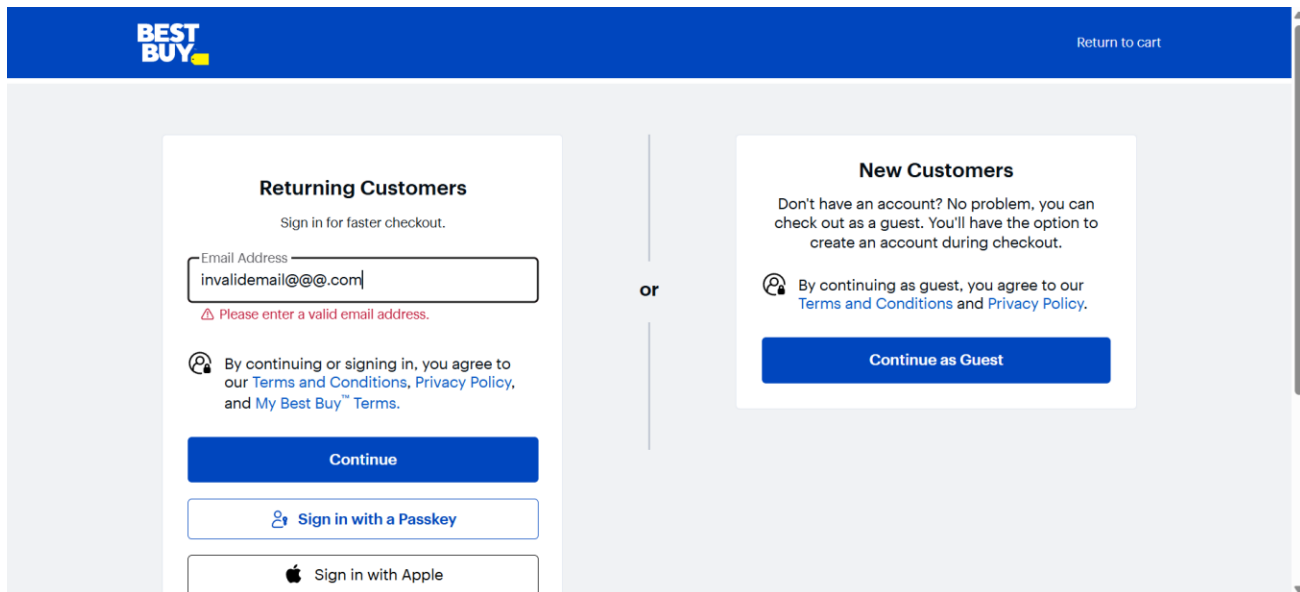


Fig-28 [BDD Allure] TC05 — Invalid Email at Checkout (Final Screenshot)

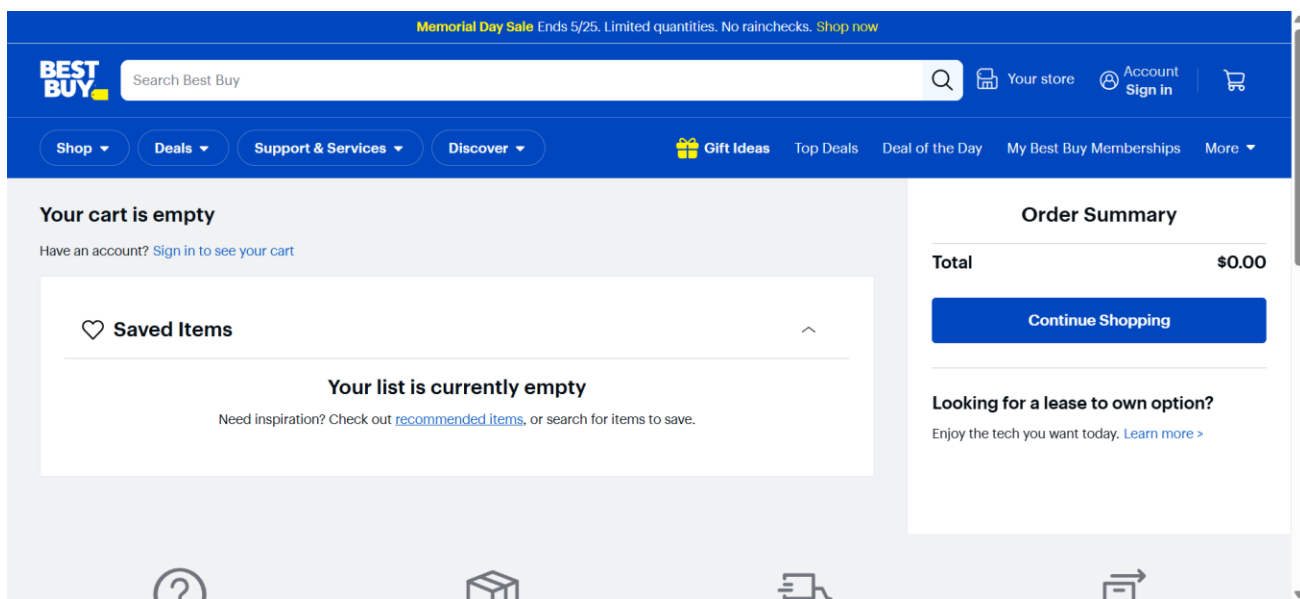


Fig-29 [BDD Allure] TC06 — Checkout Without Product (Final Screenshot)

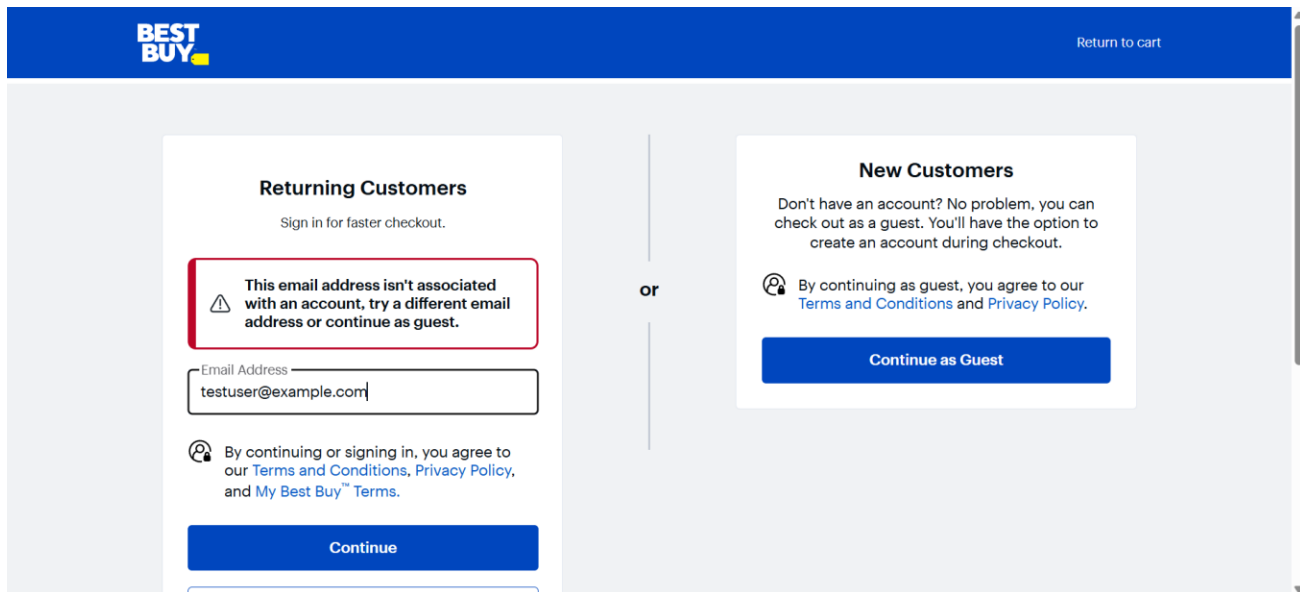


Fig-30 [BDD Allure] TC07 — End-to-End (Final Screenshot)

2.9 Test Report Summary — BDD Behave

Total Scenarios Executed	7
Scenarios Passed	7
Scenarios Failed	0
Automation Accuracy	100%

2.10 Conclusion — BDD Behave

The Selenium Behave BDD automation framework successfully validates the Best Buy Deal of the Day module using Behavior Driven Development methodology. The framework uses Gherkin Feature Files and Given-When-Then structure which improves readability, maintainability, and collaboration between technical and non-technical stakeholders. All 7 scenarios executed successfully with 100% pass rate, with professional Allure Reports, per-step and per-scenario screenshot evidence, and detailed execution logs.

PART 3 — FRAMEWORK COMPARISON

Both frameworks automate the Best Buy Deal of the Day module and achieve 100% pass rates. The table below highlights key differences:

Aspect	Selenium PyTest	Selenium BDD Behave
Test Language	Python + PyTest decorators	Gherkin (Given/When/Then)
Test File Format	.py test files with @allure annotations	.feature files + step definitions
Data-Driven	CSV via CSVReader utility	CSV via CSVReader utility
Screenshot Strategy	End-of-test via pytest hook	After every step + end of scenario
Reporting	Allure Reports with per-test logs	Allure Reports with per-scenario logs
Module Under Test	Deal of the Day — Deal of Day flow	Deal of the Day — Deal of Day flow
Total Test Cases	7 (1 E2E + 4 Positive + 2 Negative)	7 (1 E2E + 4 Positive + 2 Negative)
Pass Rate	100% (7/7)	100% (7/7)
Stakeholder Readability	Technical (Python)	Business-readable (Gherkin)
Architecture	Page Object Model	Page Object Model

Overall Conclusion

Both the Selenium PyTest and Selenium BDD Behave automation frameworks successfully validate the Best Buy Deal of the Day module with 100% automation accuracy across all 14 combined test cases. The PyTest framework excels in technical flexibility and rapid test development, while the BDD Behave framework provides superior readability through Gherkin syntax enabling collaboration with non-technical stakeholders.

Both frameworks implement the Page Object Model design pattern ensuring maintainability and reusability, integrate Allure reporting for professional test evidence, employ CSV-based data-driven testing, and capture screenshots at key execution stages. Together they represent a comprehensive, industry-standard automation testing solution for the Best Buy e-commerce platform.